

## Lab 10

### Mini- Project

In this project, we will design a 5GHz oscillator in TSMC 65nm with the following specs

| Specification                                    | Min  | Typical        | Max               |
|--------------------------------------------------|------|----------------|-------------------|
| Supply                                           | 1    | 1.1            | 1.2               |
| Temperature                                      | -40  | 27             | 125               |
| Output Clock frequency<br>(with PVT variations)  | -45% | 5 GHz          | 55%               |
| Output Clock Duty cycle<br>(with PVT variations) | -45% | 50%            | 55%               |
| Overall Current consumption                      |      | 5 mA           |                   |
| Capacitance Load on Output Clock                 |      | 10 femto farad |                   |
| Rise and fall time of Output Clock               |      |                | 5% of clock cycle |
| Input Bias current<br>(V/R poly current)         |      | 20 uA          |                   |

#### Requirements:

- Snapshots for the design showing the devices' sizes
- Snapshots for the schematic and setup for each test bench used during design.
- Typical results showing the clock waveform in the transient, measured frequency, duty cycle, rise/fall time, and average current consumption
- PVT results showing the measured frequency, duty cycle, rise/fall time, and average current consumption
- Plot the frequency versus the temperature for the PVT corners
- Plot the frequency versus supply for the PVT corners
- Plot the average current consumption versus temperature for the PVT corners
- Table of results versus the required specs

#### Simplified Schematic Overview:

